EASYCARE TO-DO APP

A Low-Latency, Voice-Enabled Offline Task-Management System

Introduction

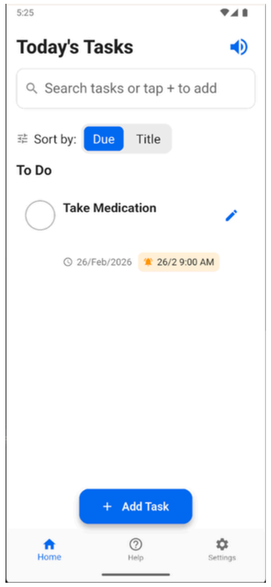
Friction in data entry hinders productivity for many users. Mainstream to-do apps rely on small text, complex menus and cloud connectivity, creating barriers for older adults. EasyCare reduces entry friction with large accessible controls and on-device voice capture so tasks can be recorded quickly and privately.

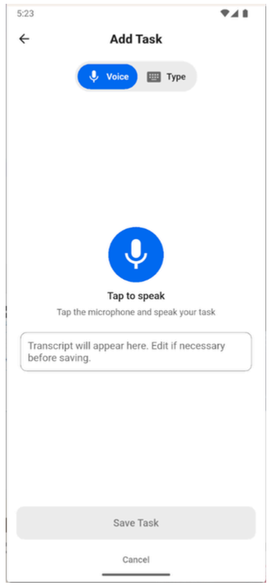
Objective

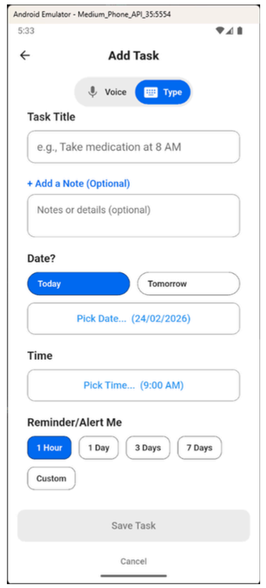
- Build an Android app with large text, high-contrast UI and large tap targets.
- Provide voice dictation with an editable transcript and TTS playback.
- Implement local timezone-aware reminders with hour presets and 1, 3 and 7 day options.
- Ensure offline persistence via SQLite(Drift).

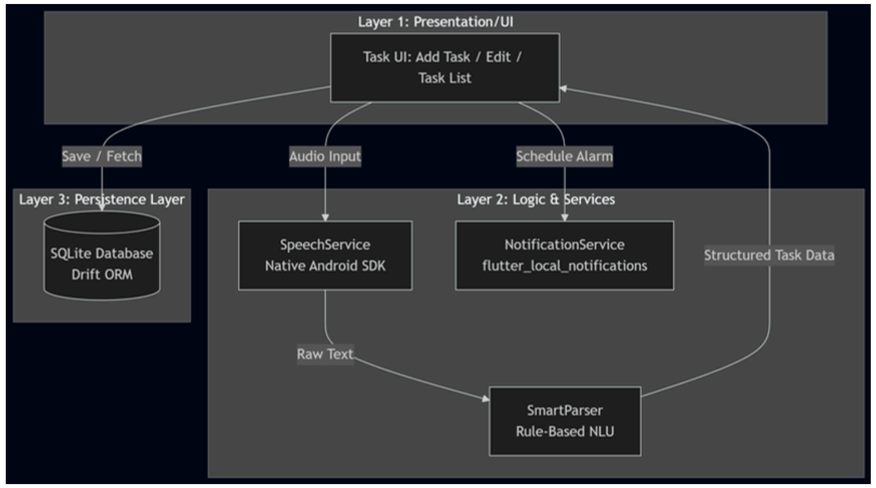
System architecture & tech stack

1. Framework: Flutter (Dart) — reactive UI, native performance
2. Persistence: SQLite via Drift — reliable offline storage and efficient sorted queries.
3. Speech engine: Android SpeechRecognizer (via speech_to_text) configured to request on-device recognition for low latency and privacy.
4. Notifications: flutter_local_notifications with timezone-aware scheduling and AndroidScheduleMode.exactAllowWhileIdle to help ensure delivery during Doze (subject to Android permissions and OEM behaviour)









Key implementation

Offline voice capture pipeline

- Uses partial results for immediate visual feedback. Silence detection (tunable pauseFor 3–5 s) signals end-of-speech. Final transcription is auto-inserted into the form for user confirmation. On-device mode reduces latency and preserves privacy.

SmartParser (rule-based NLU)

- Lightweight regex tokenizer extracts title, date, time and reminder fields. Recognises relative dates (e.g. 'tomorrow'), absolute dates and 12/24-hour time formats. Converts natural phrases like 'remind me 2 hours before' into Duration objects for the scheduler.

Features & Results

- Offline operation: full CRUD, voice capture and TTS work without network.
- Natural language entry populates Title, Date, Time and Reminder fields.
- Low-latency partial results provide immediate feedback.
- Timezone-aware scheduling persists across reboots.
- Accessible UI prioritises large text and simple controls.

Conclusion & Future work

EasyCare shows on-device speech and local parsing can reduce capture time and protect privacy.

Future directions

- Cloud sync (optional, encrypted) for multi-device support.
- Replace or augment the regex parser with a small on-device language model (SLM) for richer intent parsing.
- Offer deterministic offline STT for devices without on-device support.

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Drift

Dart